FYS5419/9419, lecture February 26, 2025

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one-qubit $\mathcal{H}$

$$\mathcal{H} = a\mathbb{1} + bZ + cX + dY$$

$$\langle \mathcal{H} \rangle = \langle \psi | \mathcal{H} | \psi \rangle$$

Basis for matrix Z, $|0\rangle \wedge |1\rangle$

$$|\psi\rangle = c_1^Z |0\rangle + c_2^Z |1\rangle$$

Basis for matrix X

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \wedge |-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$|i\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\ i\end{bmatrix}, \quad |-i\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\ -i\end{bmatrix}$$

eigenbasis for Y

$$X|+\rangle = +1\,|+\rangle$$

$$X|-\rangle = -1\,|-\rangle$$

$$Y|i\rangle = +1\,|i\rangle$$

$$\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$$

$$\langle\psi|Z|\psi\rangle =$$

$$\left((c_1^z)^*\langle 0| + (c_2^z)^*\langle 1|\right) Z \left(c_1^z|0\rangle + c_2^z|1\rangle\right)$$

$$= |c_1^z|^2 - |c_2^z|^2$$

$$\langle \psi | x | \psi \rangle$$

$$c_1^x |+\rangle + c_2^x |-\rangle$$

$$= |c_1^x|^2 - |c_2^x|^2$$

$$\langle z \rangle = (n_0 - n_1)/N$$

after N-measurements

\# of times $|c_1^z|^2$

\# of times $|c_2^z|^2$

$$\langle X \rangle = \frac{n_+ - n_-}{N}$$

initial state $|0\rangle$

$$|\psi\rangle \cdot \; R_y(\phi) R_x(\theta) |0\rangle$$

$$|0\rangle \;—\; \boxed{R_x(\theta)} \;—\; \boxed{R_y(\phi)} \;—\; |\psi\rangle$$

$, u_2 - - -$
$u_M |0\rangle$

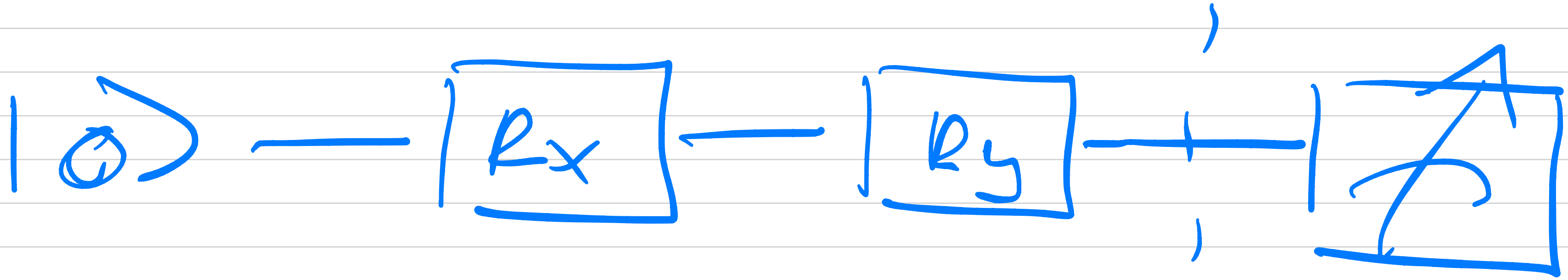
$$X = HZH$$

$$|\psi'\rangle = \underline{H}\, R_y(\phi) R_x(\theta) |0\rangle$$

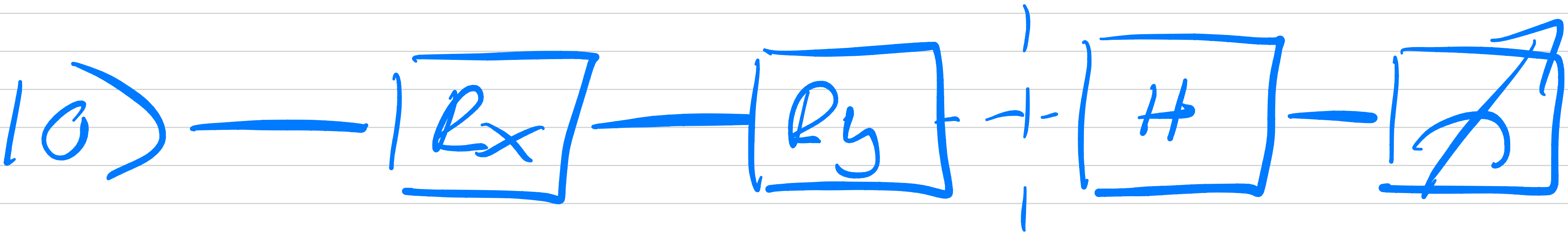
Hadamard

$$\mathcal{H} = a I + b \langle \psi | Z | \psi \rangle$$
$$+ c \langle \psi | X | \psi \rangle$$
$$= a \mathbb{1} + b \langle \psi | Z | \psi \rangle$$
$$+ c \langle \psi' | Z | \psi' \rangle$$
$$X = HZH$$
$$|\psi'\rangle = H R_y R_x |0\rangle$$

$$\langle \psi | Z | \psi \rangle$$



$$\langle \psi | X | \psi \rangle$$

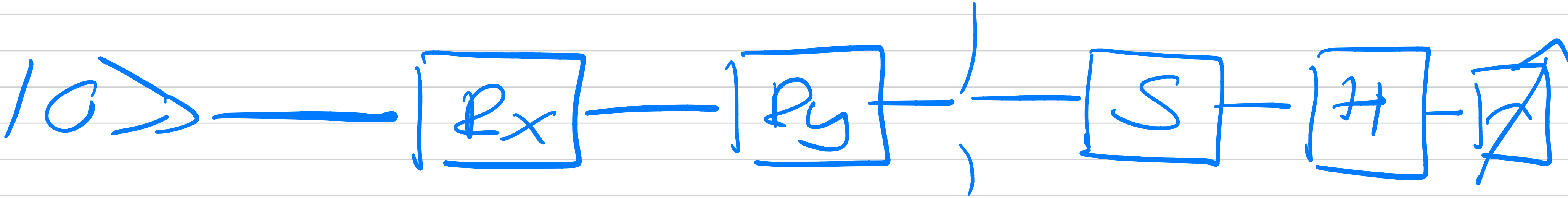


$$= \langle \psi' | Z | \psi' \rangle =$$

$$\langle \psi H | Z | H \psi \rangle$$

$$\langle \psi | Y | \psi \rangle$$

$$Y = H S^{\dagger} Z H S$$



$$\langle \mathcal{H} \rangle = \langle \psi | \mathcal{H} | \psi \rangle \geq E_0$$

$$\langle \mathcal{H}^2 \rangle = \langle \psi | \mathcal{H}^2 | \psi \rangle = E_0^2$$

if $\mathcal{H}|\psi\rangle = E_0 |\psi\rangle$

$$\mathrm{var}[\mathcal{H}] = \langle \mathcal{H}^2 \rangle - (\langle \mathcal{H} \rangle)^2 = 0$$

Two-qubit Hamiltonian

$$\mathcal{H} = \begin{bmatrix} \varepsilon_0 + H_z & 0 & H_x & \\ 0 & \varepsilon_1 - H_z & H_x & 0 \\ 0 & H_x & \varepsilon_2 - H_z & 0 \\ H_x & 0 & 0 & \varepsilon_3 + H_z \end{bmatrix}$$

$$= \mathcal{H}_0 + \mathcal{H}_I$$

$$\mathcal{H}_0 = \begin{bmatrix} \varepsilon_0 & & 0 & \\ & \varepsilon_1 & & \\ & 0 & \varepsilon_2 & \\ & & & \varepsilon_3 \end{bmatrix}$$

$\mathcal{H}_0 |00\rangle = \varepsilon_0 |00\rangle$ $\quad$ $\mathcal{H}_0 |10\rangle = \varepsilon_2 |10\rangle$

$\mathcal{H}_0 |01\rangle = \varepsilon_1 |01\rangle$ $\quad$ $\mathcal{H}_0 |11\rangle = \varepsilon_3 |11\rangle$

$$H_I = H_x \, X \otimes X + H_z \, Z \otimes Z$$

$$Z \otimes Z = \begin{bmatrix} \begin{matrix} 1 & 0 \\ 0 & -1 \end{matrix} & 0 \\ 0 & \begin{matrix} 1 & 0 \\ 0 & -1 \end{matrix} \end{bmatrix}$$

$$\underbrace{X \otimes X} = \begin{bmatrix} 0 & X_{2\times 2} \\ X_{2\times 2} & 0 \end{bmatrix}$$

$$P = U_p^{\dagger} \, M \, U_p \qquad Z$$

$$(X = HZH)$$